Forecasting Weekly 10-Year Government Bond Yield Changes Using JPMaQS Macro-Quantamental Indicators

BFI JPMaQS Macro-Quantamental Data Competition | Yun Liu | University of Chicago

1. Economic Rationale and Predictor Selection

The standard decomposition of nominal bond yields into a real rate and an inflation compensation component (the Fisher identity) motivates the three principal prediction channels used in this model: inflation expectations, monetary policy stance, and risk appetite. Weekly yield changes are inherently noisy — they blend permanent macro revisions with transient liquidity and flow effects — but point-in-time macro-quantamental indicators provide the cleanest available signal because they reflect exactly the information set available to market participants at each date, free of look-ahead revision bias.

Inflation and Real Yield Signals

Realized CPI inflation (CPIH_SA_P1M1ML12, CPIC_SA_P1M1ML12) and 1-year-ahead inflation expectations (INFE1Y_JA) are the most direct bond yield drivers: higher realized and expected inflation mechanically raises the compensation investors demand, and signals future central bank tightening. An engineered real-yield spread feature (nominal yield carry minus expected CPI) captures the valuation wedge: compressed real yields historically precede yield normalization. This is consistent with the growing empirical literature linking real rate mean-reversion to subsequent yield rises (e.g., Campbell & Shiller, 1991; Cochrane & Piazzesi, 2005).

Bond Carry and Yield Curve Shape

Bond carry — the return from holding a bond assuming an unchanged yield curve — at 2-year (DU02YCRY_NSA) and 5-year (DU05YCRY_NSA) maturities encodes current yield curve slope and curvature. Positive carry is a documented predictor of near-term bond returns across DM markets (Koijen et al., 2016). An engineered carry-slope feature (5Y carry minus 2Y carry) additionally captures the curvature of expectations about the policy rate path: a flattening carry slope often signals an impending policy pivot, creating asymmetric yield dynamics across the curve.

Risk Appetite and Cross-Asset Signals

Equity excess returns (EQXR_NSA) emerged as the single strongest predictor in feature importance analysis. This reflects the well-documented risk-on/risk-off dynamic: equity rallies reduce safe-haven demand for government bonds, pushing yields higher. The cross-sectional z-score of equity returns (EQXR_NSA_csz) captures relative risk appetite across countries, allowing the model to distinguish country-specific from global equity-bond co-movement.

Credit and Growth Conditions

Credit-to-GDP (PCREDITGDP_SA) proxies for above-potential economic expansion: rapid credit growth signals overheating, raising expected policy rates and term premia. This variable has historically led yield rises by 1–2 quarters in panel studies of DM economies.

A pooled panel of nine developed market countries (USD, EUR, GBP, JPY, AUD, CAD, CHF, SEK, NOK) was used rather than country-specific models. This exploits the common macro-yield transmission mechanism across DM economies while country dummy variables absorb persistent structural differences (e.g., JPY's chronic near-zero-rate regime, CHF's safe-haven premium).

2. Model Architecture and Training Methodology

Ensemble Design

A VotingRegressor ensemble combines three base learners with complementary inductive biases, equal-weighted. The motivation is regime robustness rather than pure CV RMSE minimization: if one model overfits to the macro regime present in the training sample, averaging dilutes its influence and prevents a single model's failure from dominating OOS predictions.

- **Ridge.** Ridge Regression (α = 194, selected by 5-fold CV): An L2-penalized linear model that preserves all predictors but shrinks correlated macro indicators toward zero jointly. Its standardized coefficients are directly interpretable as signal strengths and serve as the economic accountability layer of the ensemble.
- **ElasticNet.** ElasticNet (α, l1_ratio searched jointly): Combines L1 sparsity with L2 grouping. L1 pressure zeroes out weaker indicators and implicitly selects the most informative macro signals within the training regime. This makes ElasticNet susceptible to overfitting when the dominant macro regime shifts between training and OOS — a risk the ensemble is designed to hedge.
- **HistGradBoost.** Histogram Gradient Boosting: Captures non-linear regime interactions — for instance, inflation signals may only predict yield rises when the central bank is not constrained at the zero lower bound, and carry signals become more informative in cutting cycles. Handles missing values natively, retains all features, and learns conditional feature importance, making it more adaptive to regime shifts than sparse linear models.

Cross-Validation and Temporal Discipline

Walk-forward TimeSeriesSplit with 5 folds and a 4-week gap was applied to the pooled panel (9,486 training rows). The gap prevents leakage from macro releases whose information state spans the fold boundary. Panels were sorted by date before splitting — not by country — to ensure strictly chronological folds that respect the causal structure of macro-financial data. All hyperparameters (Ridge α, ElasticNet α and l1_ratio) were selected via nested CV on the training set only.

Feature Engineering

Three engineered features were added beyond the 30 downloaded JPMaQS indicators:

- carry_slope = DU05YCRY_NSA − DU02YCRY_NSA: captures yield curve carry steepness.
- ryld_spread = nominal 10Y yield carry minus expected CPI: real yield valuation signal.
- EQXR_NSA_csz: cross-sectional z-score of equity excess returns across the 9 countries.

All features were forward-filled across short gaps (≤ 4 weeks) to handle monthly/quarterly release schedules, then median-imputed within each pipeline to handle any residual missing values robustly. StandardScaler was applied before Ridge and ElasticNet; HistGradBoost is scale-invariant.

3. Results and Key Findings

Cross-Validation Performance

CV RMSE results (5-fold walk-forward, training set): ElasticNet 0.882%, Ensemble 0.927%, Ridge 0.959%, HistGradBoost 1.021%. ElasticNet achieved the lowest CV RMSE — the ensemble does not improve on it by this metric. Fold-level RMSE ranged from 0.741% to 1.198%, reflecting varying macro volatility across sub-periods; the highest-error fold coincides with the 2024 repricing of U.S. rate cut expectations. Critically, all five folds fall within the post-2020 hiking-and-tightening era: no fold tests model behaviour in a rate-cutting regime, which limits CV's reliability as a selection criterion across macro regimes.

Out-of-Sample Performance (52 weeks, March 2025 – March 2026)

The ensemble OOS RMSE of 0.688% is lower than its CV RMSE of 0.927%, but this does not indicate the model improved — it reflects a calmer bond market in the OOS period. OOS return volatility was 0.713% per week, well below the training-era volatility implied by CV RMSEs near 0.93%. Benchmarked correctly against the naive predict-zero baseline (RMSE = 0.714%), the ensemble reduces RMSE by 3.6%, a modest but real skill margin across 468 country-week observations.

The per-model OOS breakdown reveals a sharp reversal from CV rankings: HistGradBoost (0.688%, 3.6% skill over naive) essentially matches the ensemble and is the dominant OOS contributor, while ElasticNet (0.702%, 1.6% skill) is the weakest — despite winning CV. Ridge (0.694%, 2.7% skill) sits between them. This reversal is consistent with ElasticNet's L1 sparsity locking in a feature set optimised for the tightening regime prevalent in training, which lost predictive power when the OOS period entered a rate-cutting cycle. HistGradBoost's non-linear, full-feature structure adapted more readily to the regime shift. The ensemble validates its design rationale: by averaging ElasticNet with the more robust models, it clips ElasticNet's OOS underperformance while preserving its CV-era contributions.

OOS directional accuracy reached 57.9%, compared to 54.5% on out-of-fold training data and a 50% naive baseline — a gain of +7.9 percentage points. Directional accuracy is the more economically meaningful metric here: even a 3.6% RMSE reduction is modest in absolute terms, but a consistent directional edge above 55% translates into positive risk-adjusted returns in a long-short duration strategy at weekly rebalancing frequency. CHF (57.7%) and JPY (57.5%) show the highest country-level accuracy, consistent with those markets having more stable macro-yield transmission via safe-haven risk appetite dynamics.

Feature Importance and Economic Interpretation

Ridge coefficient magnitudes on standardized features (Figure 1) rank the predictors in order of economic signal strength. The top five — equity excess return (EQXR_NSA, 0.122), 5Y bond carry (DU05YCRY_NSA, 0.099), 2Y bond carry (DU02YCRY_NSA, 0.070), engineered carry slope (0.070), and cross-sectional equity z-score (EQXR_NSA_csz, 0.062) — all align with the economic channels described in Section 1. The top CPI predictor (CPIH_SA_P1M1ML12, 0.060) and inflation expectations (INFE1Y_JA, 0.046) round out the inflation channel. Country dummies (CHF, EUR) absorb persistent structural differences rather than acting as information-bearing predictors.

Country-Level Accuracy

CHF (57.7%) and JPY (57.5%) achieved the highest directional accuracy. Both are safe-haven currencies whose bond yields exhibit tighter links to global risk appetite — the equity-yield channel dominant in EQXR_NSA — consistent with the feature importance results. USD showed the lowest directional accuracy (52.6%), reflecting the greater idiosyncratic policy noise in U.S. Treasuries during the OOS period of recalibrating Fed rate expectations.

Model Limitations

Three limitations are worth noting. First, out-of-fold residuals show a modest negative mean across all countries (−0.12 to −0.35% per week): the model systematically under-predicts return magnitudes, a known consequence of L2/L1 shrinkage compressing predictions toward zero. A post-hoc scaling correction or asymmetric loss function could partially address this. Second, walk-forward CV folds all fall within the post-2020 tightening era; no fold evaluates the model in a cutting-cycle regime, which explains why CV rankings (favouring ElasticNet's tightening-regime coefficients) diverge from OOS rankings (favouring HistGradBoost's regime-adaptive non-linearity). Future work should include explicit regime-stratified evaluation. Third, the model uses a static, pooled feature set; a regime-switching or time-varying weight extension that up-weights recent observations during macro transitions could further improve robustness to structural breaks.